

Question number	Answer	Additional guidance	Mark
6(a)	B is the correct answer A is not the correct answer as auxin would not increase the likelihood that a neuron will produce a nerve impulse at the post synaptic membrane as it is a plant hormone C is not the correct answer as gibberellin would not increase the likelihood that a neuron will produce a nerve impulse at the post synaptic membrane as it is a plant hormone D is not the correct answer as statin would not increase the likelihood that a neuron will produce a nerve impulse at the post synaptic membrane as it is a drug used to reduce cholesterol		(1)

Question number	Answer	Additional guidance	Mark
6(b)(i)	An answer that includes three of the following points: • low oxygen concentration decreases cell survival (1) • as lidocaine concentration increases {cell survival / percentage} increases (1) • error bars do not overlap so significantly different (1)	ACCEPT converse argument / description ACCEPT with lidocaine cell survival does not reach {100% / as high / doesn't increase as much} (as control does) / control has the highest survival ACCEPT lidocaine in 0.01 has almost same effect as oxygen rich cells ACCEPT positive correlation ACCEPT error bars do overlap for control and 0.01 so results not significantly different	(3)

Question number	Answer	Additional guidance	Mark
6(b)(ii)	B is the correct answer A is not the correct answer as it is not a hormone that initiates differentiation C is not the correct answer as it is not how a heart beat is initiated in the atrio-ventricular node D is not the correct answer as it is not how muscles work in pairs		(1)

Question number	Answer	Additional guidance	Mark
6(b)(iii)	An answer that includes two of the following points: • {prevents / reduces / inhibits} {(electrical)impulses / action potentials / depolarisation} (spreading over heart muscle) (1) • lidocaine reduces {rate of contraction / (heart) muscle cell contraction} (1) • (therefore) reducing the need for oxygen (for respiration) / reducing need for ATP (1)	ACCEPT prevent diffusion of sodium ions / prevents influx of sodium ions ACCEPT low(er) {heart rate / stroke volume} ACCEPT resulting in reduced conversion of pyruvate into lactate / {no / reduced} lactic acid	(2)

Question Number	Answer	
* 6(c)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant.	(6)
1	Consideration of heart rate / bpm and exercise (Graph 1). • At start of exercise HR increases in both athletes and non athletes • the increase in HR is steeper in non athletes than athletes • peak HR higher in non athletes than athletes • HR levels off in both athletes and non athletes during exercise • at end of exercise HR reduces faster in athletes than non-athletes • at end of exercise HR returns to normal faster in athletes and non athletes	
2	Consideration of weekly physical activity and mortality risk (Graph 2). • Person with CVD and no physical activity per week risk of death is greatest • risk of CVD reduces with increase in physical activity per week for both person with or without CVD • if no CVD there is a lower risk of dying than if you have CVD • risk of death decreases more in person with CVD than without as duration of exercise increases. • If person with CVD increases duration of exercise the risk of death almost meets the same level as a person without CVD	
3+4	Consideration and explanation of health benefits to improve the outcomes for patients with CVD Benefits Heart / Blood circulation • exercise leads to a stronger heart as {cardiac muscle increases / increase in size of heart chambers} • Leading to more oxygenation of cells / heart pumps more efficiently/higher stroke volume /higher cardiac output • exercise can lead to reduce blood pressure due to {greater flexibility of blood vessels /vasodilation / • exercise improves {lung function/ lung capacity / lungs} / as more oxygen taken up in the alveoli • exercise increases volume of blood circulating as blood vessels {less stiff / more flexible} • exercise stimulates the release of Nitric Oxide leading to vasodilation of blood vessels	
5	CVD • reduced {blood pressure / heart rate / cholesterol} linked to reduced risk of atheroma / plaques	

6 7	- athletes have a reduced risk of atherosclerosis - exercise can reduce plaques leading to improved blood circulation Hormones / Brain / Metabolism - exercise reduce levels of stress hormones eg cortisol, adrenaline which positively enhances body function / metabolism - exercise causes release of 'feel good' hormones which reduces nervous transmission within the brain / reduces stress - exercise leads to cells becoming more responsive to insulin so cells {absorb / metabolise / respire} more glucose / increase weight loss so less strain on heart - exercise reduce inflammation throughout the body / reduces CRP levels - exercise leads to enhanced mitochondrial function leading to greater {respiration / heart contraction} - exercise leads to greater stores of {myoglobin / oxygen} in muscles so more aerobic respiration Conclusion / summary / comment on methodology No detail about age / gender / lifestyle / numbers involved However other factors may also contribute to death by CVD – lifestyle, diet, genetics, smoking, drinking, obesity	
	Additional guidance	
Level 0	0	No awardable content
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and / or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and / or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.

